SARTHAK PAINULI

Software Engineer
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Summary

Software Engineer with ~1 year of professional experience and 1 year as an intern, working across the full stack from payment APIs to the screens built on them. I like getting to the bottom of why things fail under real conditions, and I don't call a feature done until it holds up under load. I pick up unfamiliar stacks quickly, having shipped production code in Haskell, Java, Python, and TypeScript.

Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, Haskell
Frameworks: FastAPI, Spring Boot, Node.js, Django, Flask
Frontend: React, Next.js, HTML, CSS, JSP
Backend & APIs: REST APIs, JWT Authentication, JDBC, PostgreSQL, MongoDB, Redis, Celery, Alembic, OpenTelemetry (OTLP)
AI & LLMs: LLM API Integration, Model Context Protocol (MCP), Prompt Engineering, RAG, LangChain, Vector Databases (Qdrant)
DevOps & Tools: Git, Linux, Docker, Docker Compose, Nginx, Postman, AWS, GitHub Actions (CI/CD)

Experience

Product Engineer - Juspay 08/2025 – 06/2026
Bengaluru, India

- Enabled Google Wallet and Apple Pay checkout for HSBC and 3+ more merchants by building a config-driven payment screen and the API data extraction behind it, as part of a multi-payment-method redesign.
- Supported 100K+ daily transactions across 3 payment providers by owning 6+ transaction flows end to end: initiation, 3DS authentication, refunds, captures, and webhook ingestion.
- Improved UI rendering performance by 15% by optimizing Presto-based modules, speeding up load times across the payment flow.
- Built a proof-of-concept MCP payment copilot for IndiGo and Air India that loads in an iframe on "Proceed to Payment" inside the airline's chatbot, letting customers pay in-chat without switching tabs.

Web Development Intern - NXP Semiconductors 07/2024 – 06/2025
Noida, India

- Replaced manual workflows across multiple teams with 8+ internal tools built in Java, JSP, HTML, CSS, JavaScript, JDBC, and SQL, gathering requirements directly from stakeholders.
- Enabled real-time tracking and reservation of 100K+ lab tools by designing an inventory and booking system with indexed availability lookups and double-booking prevention.
- Grew usage of an internal dev tooling platform by 40% by reworking core workflows and fixing reliability issues blocking adoption.

Projects

Ghost Protocol github.com/sarthak310104/Ghost-Protocol-emp Live Demo: ghost-protocol-emp.vercel.app
Python, FastAPI, Next.js/TypeScript, PostgreSQL, Redis, Celery 09/2026 – Present

- Built a self-hosted observability platform that flags anomalies and bottlenecks before they cause incidents, by streaming services' OpenTelemetry traces into a FastAPI/AsyncIO backend that maps them as a behavioral graph.
- Sustained 11k-23k spans/sec with API p95 under 14ms on a 50-service synthetic topology by load-testing ingestion and fixing a Postgres deadlock (silently dropping ~25% of graph updates) and a follow-on write race.
- Projected incident recovery with 95% confidence intervals by building two evidence engines: a mean-reversion simulator and a concurrent-cohort comparator for live canary/rollout traffic splits.
- Surfaced config changes up to 200 deploys before an incident as evidence by tracking per-key value changes over time.

Control-Tower github.com/sarthak310104/Control-Tower Live Demo: control-tower-lv5k.onrender.com/ui
Python, FastAPI, PostgreSQL, MCP, tree-sitter, Alembic, Three.js, Docker 08/2026 – Present

- Prevented AI coding agents (e.g. Claude Code and Codex) sharing a codebase from overwriting each other by building a coordination service where agents claim the functions they edit through an 11-tool MCP server or REST API.
- Supported function-level claims in 5 languages by parsing code with Python's AST and tree-sitter (JavaScript, TypeScript, Go, Java), keying claims on symbol names so they stay valid as code shifts.
- Guaranteed exactly one grant out of 12 simultaneous claims on one function by adding per-file Postgres advisory locks, and caught unclaimed edits by reconciling each agent's git diff against its claims.
- Verified per-agent token auth, migrations, and the MCP server end to end with 96 tests in CI, plus a 3D Three.js dashboard.

Education

B.Tech in Computer Science Engineering 11/2021 – 06/2025
Jaypee Institute of Information Technology

Achievements

- 3-star CodeChef, 1722 peak rating
- 200+ LeetCode problems solved

Socials

LeetCode: leetcode.com/u/sarthak310104 | CodeChef: codechef.com/users/galahad_3101
HackerRank: hackerrank.com/profile/sarthakpainuli11 | Portfolio: sarthak-painuli-portfolio.vercel.app